Relational Geometry and the Emergence of Dimensions

A Minimal Generative Algebraic Model

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Status. This manuscript proposes an exploratory generative framework. It is *not* presented as a complete physical theory. Its aim is to define a minimal internal formalism, clearly separate postulates from derived relations, and distinguish algebraic results from programmatic physical correspondences.

Prologue: On the Spirit of This Inquiry

Many physical explanations are first learned as labels: light comes "by light," magnets attract "by magnetism," bodies fall "by gravity." Labels predict, but they do not necessarily reveal what is being measured.

One question remains resistant to isolation: the present we inhabit. If time is defined by periodic comparison, perhaps it is not primitive, but a structure that appears once relations become stable enough to be compared.

The ancient Chinese philosopher Zhuangzi captured this journey from undifferentiated unity to fragmented multiplicity with remarkable precision: *"El conocimiento de los antiguos era perfecto. ¿Cómo era perfecto? Al principio, todavía no sabían que había cosas. Este es el conocimiento más perfecto, porque no se le puede agregar nada. Luego, sabían que había cosas, pero no hacían distinciones entre ellas. Luego hicieron distinciones, pero aún no emitían juicios sobre ellas. Cuando surgieron los juicios, el Todo fue destruido. Con la destrucción del Todo, surgió el sesgo individual."*

This work asks: what minimal algebraic structure must exist for this process of distinction and fragmentation to become coherent? A normalized reference $\hat{\aleph}$ encodes the primordial state where "no sabían que había cosas." A differentiation marker $\boldsymbol{R}$ introduces the first distinction. A generative operator $\times$ produces the subsequent layers of differentiation, until identity and orientation emerge – the "sesgo individual."

Abstract

We define a minimal relational algebra in which new degrees of freedom are generated by successive projections relative to a distinguished direction. Starting from a normalized reference $\hat{\aleph}$ and a differentiation marker $\boldsymbol{R}$ with $\|\boldsymbol{R}\| = R \in [0, 1]$, we postulate an antisymmetric bilinear operator $\times$ satisfying restricted norm compatibility (only relative to $\boldsymbol{R}$), metric compatibility in the orthogonal subspace, and a double-projection closure rule with binary sign $\sigma \in \{\pm 1\}$. From these assumptions, a hierarchy $(\boldsymbol{U}, \boldsymbol{V}, \boldsymbol{m}, \boldsymbol{a})$ emerges with controlled scaling in R. A binary closure depth n is introduced and interpreted representation-theoretically. For spin-$\frac{1}{2}$, the minimal real representation dimension gives $n = 4$. A proton-scale consistency target follows as a falsifiable consequence. Further tentative correspondences (valence heuristics, Hubble analogy, weak-field gravity) are presented as program notes, explicitly separated from definitive

derivations. The presentation is explicitly exploratory; all physical correspondences are labeled as programmatic and falsifiable targets rather than definitive claims.

1 Algebraic Foundations

1.1 Primitive elements

Let V be a real normed vector space.

Postulate 1 (Reference normalization)**.**

$$\left\|\hat{\aleph}\right\| = 1, \qquad \|\boldsymbol{R}\| = R \in [0,1].$$

Postulate 2 (Reference orthogonality)**.**

$$\left\langle \hat{\aleph}, \boldsymbol{R} \right\rangle = 0.$$

Define

$$V_R := \{A \in V \mid \langle A, \boldsymbol{R}\rangle = 0\}.$$

1.2 Generative operator $\times$

Postulate 3 (Bilinearity and antisymmetry)**.** *For all $A, B, C \in V$ and scalars α, β,*

$$(\alpha A + \beta B) \times C = \alpha(A \times C) + \beta(B \times C),$$

$$A \times B = -(B \times A).$$

Postulate 4 (Restricted norm compatibility)**.** *For all $A \in V_R$,*

$$\|A \times \boldsymbol{R}\| = \|A\| \, \|\boldsymbol{R}\| \, .$$

Postulate 5 (Metric compatibility in V_R)**.** *For all $A \in V_R$,*

$$\langle A \times \boldsymbol{R}, A\rangle = 0.$$

Postulate 6 (Double-projection closure)**.** *For all $A \in V_R$,*

$$(A \times \boldsymbol{R}) \times \boldsymbol{R} = \sigma R^2 A, \qquad \sigma \in \{\pm 1\}.$$

Remark 1 (Algebraic scope)**.** *Requiring $\|A \times B\| = \|A\| \, \|B\|$ for all A, B would force a normed composition algebra (Hurwitz theorem, 1898), which exists only in dimensions $1, 2, 4, 8$. Here compatibility is imposed only relative to $\boldsymbol{R}$, consistent with cross-product geometry in $\mathbb{R}^3$ and the adjoint action in $\mathfrak{su}(2)$. Thus the present framework does not assume a full composition algebra structure.*

Remark 2 (On σ)**.** *The sign σ is a discrete degree of freedom attached to the double-projection. Later, when mapping to physical interpretation, it may play a spin-like role as a binary orientation class (e.g., distinguishing particle from antiparticle, or two helicity states). In the internal formalism it is simply a label distinguishing two closure types.*

2 Generated Directions

2.1 First projection: space of differences

Definition 1.

$$\boldsymbol{U} := \hat{\aleph} \times \boldsymbol{R}.$$

Since $\hat{\aleph} \in V_R$,

$$\|\boldsymbol{U}\| = R.$$

By metric compatibility,

$$\left\langle \boldsymbol{U}, \hat{\aleph} \right\rangle = 0, \qquad \langle \boldsymbol{U}, \boldsymbol{R} \rangle = 0.$$

$\boldsymbol{U}$ represents the first emergent informational degree of freedom – an "arena of distinguishability."

2.2 Second projection: transport/orientation mode

Definition 2.

$$\boldsymbol{V} := \boldsymbol{U} \times \boldsymbol{R}.$$

Using Postulate 6 with $A = \hat{\aleph}$,

$$\boldsymbol{V} = (\hat{\aleph} \times \boldsymbol{R}) \times \boldsymbol{R} = \sigma R^2 \hat{\aleph}, \qquad \|\boldsymbol{V}\| = R^2. \tag{1}$$

The magnitude is independent of σ; the direction encodes the orientation class.

2.3 Third projection: confinement/identity mode

We impose an orthogonal decomposition that separates the generated difference into a transport-like part and an orthogonal residual.

Postulate 7 (Orthogonal decomposition)**.** *There exists $\boldsymbol{m} \in V_R$ such that*

$$\boldsymbol{U} = \boldsymbol{V} + \boldsymbol{m}, \qquad \langle \boldsymbol{m}, \boldsymbol{V} \rangle = 0.$$

Then, by Pythagoras,

$$\|\boldsymbol{m}\|^2 = \|\boldsymbol{U}\|^2 - \|\boldsymbol{V}\|^2 = R^2 - R^4, \qquad \Rightarrow \qquad \|\boldsymbol{m}\| = R\sqrt{1 - R^2}.$$

Remark 3 (Extrema)**.** $\|\boldsymbol{m}\| = 0$ *at* $R = 0$ *and* $R = 1$*, and attains its maximum at* $R = 1/\sqrt{2} \approx 0.707$*.*

2.4 Fourth projection: interaction/adjustment mode

Definition 3.

$$\boldsymbol{a} := \boldsymbol{V} \times \boldsymbol{R}.$$

Since $\boldsymbol{V} \in V_R$,

$$\|\boldsymbol{a}\| = R^3.$$

This mode encodes a need for reconfiguration when relational modes interact – the seed of what becomes force-like behavior after calibration.

2.5 Hierarchy summary

$$\|\boldsymbol{U}\| = R, \quad \|\boldsymbol{V}\| = R^2, \quad \|\boldsymbol{m}\| = R\sqrt{1 - R^2}, \quad \|\boldsymbol{a}\| = R^3.$$

3 Binary Closure Depth

After calibration (using c and $\hbar$ as anchors), norms are interpreted as frequencies. The internal relation

$$\|\boldsymbol{U}\|^2 = \|\boldsymbol{V}\|^2 + \|\boldsymbol{m}\|^2$$

becomes

$$\omega_N^2 = \omega_V^2 + \omega_m^2. \tag{2}$$

Postulate 8 (Binary closure)**.** *A maximally stable family satisfies*

$$\frac{\omega_V^2}{\omega_N^2} = 2^{-n}, \qquad n \in \mathbb{N},$$

equivalently $\omega_V^2 : \omega_m^2 = 1 : (2^n - 1)$.

Remark 4 (Structural meaning of $n = 4$ (spin-$\frac{1}{2}$ motivation))**.** *This manuscript does not claim* a theorem *fixing n. We only record a structural motivation: for a spin-$\frac{1}{2}$ system the observable algebra is generated by Pauli matrices, satisfying*

$$\sigma_i \sigma_j = \delta_{ij} I + i\epsilon_{ijk}\sigma_k, \qquad (\mathbf{n} \cdot \boldsymbol{\sigma})^2 = I,$$

with $\mathbf{n}$ *a unit vector. The latter identity is analogous to the return rule Postulate 6 (see also Equation* (1)*) under a reading where* $\boldsymbol{R}$ *plays the role of a measurement axis.*

The fundamental representation of $\mathfrak{su}(2)$ *is* 2*-dimensional over* $\mathbb{C}$*, hence* 4*-dimensional over* $\mathbb{R}$*. In the present program, n is interpreted as an* encoding depth *measuring the minimal independent real degrees needed to store a coherent non-commutative spin description alongside a reference state, a measurement axis, a non-commuting generator, and a binary orientation label. Under that reading, the baryonic assignment $n = 4$ is the first depth compatible with this minimal real dimensionality. A complete selection rule remains open.*

4 Proton-Scale Consistency Target

Let

$$\omega_p = \frac{m_p c^2}{\hbar}, \qquad \bar{\lambda}_p = \frac{\hbar}{m_p c}.$$

For $n = 4$,

$$\omega_V = \frac{\omega_p}{4}.$$

Define the associated length scale

$$R_V := \frac{c}{\omega_V} = 4\bar{\lambda}_p. \tag{3}$$

Using CODATA 2018 values,

$$R_V \approx 0.8412356\,\text{fm}.$$

Remark 5 (Falsifiability statement)**.** *This numerical proximity to proton charge-radius determinations (CODATA 2018:* 0.8414(19) fm*, muonic hydrogen:* 0.84087(39) fm*) is presented as a* target *of the $n = 4$ instance, not as confirmation. If a consolidated future determination of r_p stabilizes outside compatibility with Equation* (3) *(and if R_V is taken to correspond to the charge radius in that mapping), then the $n = 4$ closure instance is ruled out or must be reinterpreted.*

5 Program Notes (Indications Across Scales)

Remark 6 (Status of this section)**.** *The items below are not derived from the algebraic framework alone; they require additional interpretative steps and are presented as heuristic connections that motivate further investigation. Each is explicitly falsifiable and serves as a target for refinement.*

5.1 Atomic scale: valence heuristic (qualitative)

A proposed heuristic uses reduced ratios of valence occupations $(N_s : N_p) \to (a : b)$ and a weight $P = a + b$, where small-P structure correlates with inert/reactive patterns in periods 2–3. A derivation from the closure postulates remains open.

5.2 Cosmological scale: Hubble analogy (structural)

The internal scaling $R \mapsto R^2$ in the transport channel suggests a structural analogy with $V = H_0 D$ under a suitable calibration. A dynamical model is required before any physical claim is made.

5.3 Gravitational scale: weak-field Poisson limit (hypothesis)

One hypothesis introduces an effective relational density $\rho_{\text{rel}} := \kappa\, \omega_m^2\, \rho_m$ and proposes a Poisson-type limit

$$\nabla^2 \phi = 4\pi\alpha\, \kappa\, \omega_m^2\, \rho_m, \qquad \Rightarrow \qquad G = \alpha\, \kappa\, \omega_m^2.$$

A key falsifiable target is a small composition dependence (WEP deviations) if $\kappa\omega_m^2$ varies with nuclear binding. Estimating the scale is an explicit next step.

6 Discussion and Open Questions

6.1 What this framework is and is not

This manuscript is a minimal generative formalism plus a calibration hypothesis. It does not yet provide: (i) a covariant dynamics, (ii) a derivation of the calibration anchors, (iii) a theorem fixing the binary depths. Its value is in conceptual economy, explicit separation of postulates vs. derived relations, and a compact list of falsifiable targets.

6.2 Open questions

1. Can the return rule and restricted norm compatibility be represented as a known algebraic structure (e.g. a Lie-type representation) without contradiction? (See Remark 1.)

2. Can the binary depth n be derived as a counting theorem of independent relational constraints (beyond the structural motivation in Remark 4)?

3. What is the minimal covariant dynamics compatible with the generative operator $\times$?

4. What empirical signature most cleanly distinguishes this program from reparameterizations of known physics (e.g. composition-dependent WEP tests)?

Conclusion

We constructed a minimal relational-generative algebra in which new degrees of freedom arise by successive projections. A binary orientation σ emerges through double projection. An orthogonal decomposition yields a confinement-like mode with norm $R\sqrt{1 - R^2}$ and a natural hierarchy of

scalings. A binary closure hypothesis discretizes stable families by depth n, and we provided a structural rationale for the baryonic choice $n = 4$ based on the non-commutative structure of spin and the real dimension of the fundamental representation of $\mathfrak{su}(2)$. As a concrete consistency check, the proton Compton frequency combined with the $n = 4$ squared-channel ratio $1 : 15$ yields a length scale $R_V = 4\bar{\lambda}_p \approx 0.8412\,\mathrm{fm}$, close to the experimental charge radius. Further cross-scale correspondences are recorded as program notes, explicitly posed as targets for refinement and falsification.

Future work includes: (i) classification of admissible realizations satisfying the restricted norm and metric compatibility, (ii) a genuine selection rule for the binary depth (beyond the structural spin motivation), and (iii) a dynamical extension of the generative operator.

Epilogue

In a point without references, where everything continues without "happening," perhaps the present is not something we inhabit – but something we *are*.

And the dimensions we discover are not containers we move through, but spaces we create – each time information needs a place to be.

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A Summary of Objects and Their Interpretations

Object	Internal meaning	Possible physical analog / interpretation
$\hat{\aleph}$	Normalized reference state	"Unison present"; global phase; vacuum-like reference
$\boldsymbol{R}$	Differentiation marker	Measurement axis; direction of comparison
$\boldsymbol{U} = \hat{\aleph} \times \boldsymbol{R}$	First projection: space of differences	Non-commuting generator; arena of distinguishability
σ	Binary orientation in double projection	Spin orientation; helicity; particle–antiparticle class
$\boldsymbol{V} = \boldsymbol{U} \times \boldsymbol{R}$	Second projection: transport mode	Relational velocity; momentum-like channel
$\boldsymbol{m}$	Third projection: confinement/identity mode	Mass/inertia proxy; self-identity channel
$\boldsymbol{a} = \boldsymbol{V} \times \boldsymbol{R}$	Fourth projection: interaction mode	Force-like / adjustment channel

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3. W. Pauli (1927). "Zur Quantenmechanik des magnetischen Elektrons." *Zeitschrift für Physik* 43, 601–623.

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1. R. Pohl et al. (2010). "The size of the proton." *Nature* 466, 213–216.

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Program Preprints and Supporting Notes (Non-validated)

The following items document the broader research program. They are cited for traceability, not as independent validation.

1. L. D. Mata Sánchez (2025). *Geometric Resonance and Binary Information Closure: A First-Principles Derivation of the Proton Charge Radius.* Zenodo. doi:10.5281/zenodo.17807496.

2. L. D. Mata Sánchez (2025). *Cosmological Dualities: Hubble Expansion and Black Hole Singularities as Geometric Limits of a Relational Projection.* Zenodo. doi:10.5281/zenodo.17808981.

3. L. D. Mata Sánchez (2026). *Gravity as Relational Difference Elimination: A Harmonic Closure Derivation.* Preprint.

Conceptual and Philosophical Influences (Non-technical)

The following works influenced conceptual framing but are not used as technical foundations of the algebra.

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2. J. L. Borges (1944). "La Biblioteca de Babel." In: *Ficciones.* Editorial Sur.

3. Zhuangzi (c. siglo IV a.C.). *Zhuangzi*, Capítulo 2, "Qiwu Lun" (Sobre la igualdad de las cosas). [La cita sobre la perfección del conocimiento antiguo puede consultarse en numerosas traducciones académicas; por ejemplo, en Burton Watson (1968), *The Complete Works of Chuang Tzu*, Columbia University Press.]

Methodological Note on Computational Assistance

The development of this manuscript involved iterative structural refinement assisted by large language models. These systems were used as tools for consistency checking, algebraic stress-testing, and language revision. They are acknowledged as methodological instruments rather than epistemic authorities. Specific tools employed include OpenAI ChatGPT, DeepSeek AI, and Google Gemini; community feedback from r/LLMPhysics also contributed to refinements.